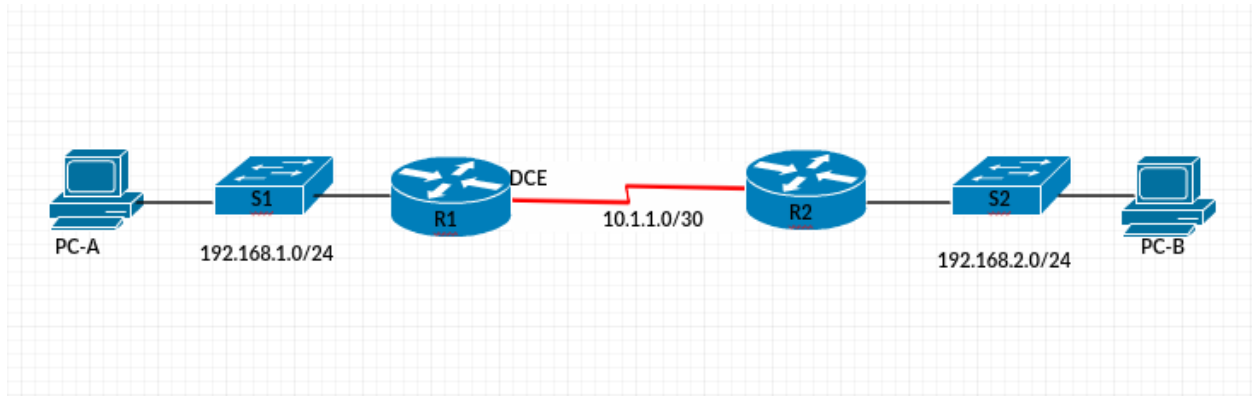


5CS031 Network Security

Workshop 1: Securing the Router for Administrative Access

Topology



IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A
	G0/0	192.168.2.1	255.255.255.0	N/A
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.2.2	255.255.255.0	192.168.2.1

Objectives

Part 1: Configure Basic Device Settings

- Cable the network as shown in the topology.
- Configure basic IP addressing for routers and PCs.
- Configure RIP routing.
- Configure PC hosts.
- Verify connectivity between hosts and routers.

Part 2: Control Administrative Access for Routers

- Configure and encrypt all passwords.
- Configure a login warning banner.
- Configure enhanced username password security.
- Configure an SSH server on a router.
- Configure an SSH client and verify connectivity.

Background / Scenario

The router is a critical component in any network. It controls the movement of data into and out of the network and between devices within the network. It is particularly important to protect network routers because the failure of a routing device could make sections of the network, or the entire network, inaccessible. Controlling access to routers and enabling reporting on routers is critical to network security and should be part of a comprehensive security policy.

In this workshop, you will build a multi-router network and configure the routers and hosts. Use various CLI tools to secure local and remote access to the routers, analyze potential vulnerabilities, and take steps to mitigate them.

Note: Before you begin, ensure that the routers and the switches have been erased and have no startup configurations.

Part 1: Configure Basic Device Settings

In Part 1, set up the network topology and configure basic settings, such as interface IP addresses.

Step 1: Cable the network.

Attach the devices, as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each router.

- Configure host names as shown in the topology.
- Configure interface IP addresses as shown in the IP Addressing Table.
- Configure a clock rate for router with a DCE serial cable attached to their serial interface. R1 is shown here as an example.

```
R1(config)# interface S0/0/0
```

```
R1(config-if)# clock rate 64000
```

- To prevent the router from attempting to translate incorrectly entered commands as though they were host names, disable DNS lookup. R1 is shown here as an example.

```
R1(config)# no ip domain-lookup
```

Step 3: Configure RIP routing on the routers.

- Use the **router RIP** command in global configuration mode to enable RIP on R1.

R1(config)# **router RIP**

- b. Configure the **network** statements for the networks on R1.

R1(config-router)# **network 192.168.1.0**

R1(config-router)# **network 10.1.1.0**

- c. Configure the hostname, IP addresses and RIP on R2.

Step 4: Verify routing information.

- a. Issue the **show ip route** command to verify that all networks display in the routing table on all routers.

R1# **show ip route**

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/3/0
L       10.1.1.1/32 is directly connected, Serial0/3/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R       192.168.2.0/24 [120/1] via 10.1.1.2, 00:00:24, Serial0/3/0
```

Step 5: Configure PC host IP settings.

Configure a static IP address, subnet mask, and default gateway for PC-A and PC-B as shown in the IP Addressing Table.

Step 6: Verify connectivity between PC-A and PC-B.

- a. Ping from R1 to R2.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

- b. Ping from PC-A, on the R1 LAN, to PC-B, on the R2 LAN.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

Note: If you can ping from PC-A to PC-B you have demonstrated that RIP routing is configured and functioning correctly. If you cannot ping but the device interfaces are up and IP addresses are correct, use the **show run**, **show ip route** commands to help identify routing protocol-related problems.

You need to complete this step before proceeding with part 2

Part 2: Control Administrative Access for Routers

In Part 2, you will:

- Configure and encrypt passwords.
- Configure a login warning banner.
- Configure enhanced username password security.
- Configure enhanced virtual login security.
- Configure an SSH server on R1.
- Research terminal emulation client software and configure the SSH client.

Note: Perform all tasks on both R1 and R2. The procedures and output for R1 are shown here.

Task 1: Configure and Encrypt Passwords on Routers R1 and R2.

Step 1: Configure a minimum password length for all router passwords.

Use the **security passwords** command to set a minimum password length of 10 characters.

```
R1(config)# security passwords min-length 10
```

Step 2: Configure the enable secret password.

Configure the enable secret encrypted password on both routers.

```
R1(config)# enable secret cisco12345
```

How does configuring an enable secret password help protect a router from being compromised by an attack?

[\[Your Answer Here\]](#)

Configuring an enable secret password help protect a router from being compromised by an attack by adding layer of security to access privileged mode using hashing .

Step 3: Configure basic console and virtual access lines.

Note: Passwords in this task are set to a minimum of 10 characters but are relatively simple for the benefit of performing the lab. More complex passwords are recommended in a production network.

- a. Configure a console password and enable login for routers. For additional security, the **exec-timeout** command causes the line to log out after 5 minutes of inactivity. The **logging synchronous** command prevents console messages from interrupting command entry.

Note: To avoid repetitive logins during this lab, the **exec-timeout** command can be set to 0 0, which prevents it from expiring. However, this is not considered a good security practice.

```
R1(config)# line console 0
R1(config-line)# password ciscocon
R1(config-line)# exec-timeout 5 0
R1(config-line)# login
R1(config-line)# logging synchronous
```

When you configured the password for the console line, what message was displayed?

[\[Your Answer Here\]](#)

```
-----
R1(config-line)#password ciscocon
% Password too short - must be at least 10 characters. Password not configured.
R1(config-line)#
```

b. Configure a new password of **ciscoconpass** for the console.

c. Telnet from R2 to R1.

```
R2> telnet 10.1.1.1
```

Were you able to login? Explain.

[\[Your Answer Here\]](#)

No, the login failed because vty lines didn't have password and login configured.

What messages were displayed?

[\[Your Answer Here\]](#)

```
R2>telnet 10.1.1.1
Trying 10.1.1.1 ...Open

[Connection to 10.1.1.1 closed by foreign host]
R2>
```

d. Configure the password on the virtual access lines (vty lines) for router R1 to allow remote access

```
R1(config)# line vty 0 4
```

```
R1(config-line)# password ciscovtypass
R1(config-line)# exec-timeout 5 0
R1(config-line)# transport input telnet
R1(config-line)# login
```

Telnet from R2 to R1 again. Were you able to login this time?

[\[Your Answer Here\]](#)

```
R2>telnet 10.1.1.1
Trying 10.1.1.1 ...Open
```

User Access Verification

Password:

R1>

Yes.

- e. Enter privileged EXEC mode and issue the **show run** command. Can you read the enable secret password? Explain.

[\[Your Answer Here\]](#)

```
R1#show run
Building configuration...

Current configuration : 1141 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
security passwords min-length 10
!
hostname R1
!
!
!
enable secret 5 $1$mERr$WvpW0n5HghRrqnrwXCUU1.
.
```

We cannot read the enable secret password as it uses hashing to encrypt password.

Can you read the console, aux, and vty passwords? Explain.

[Your Answer Here]

```
line con 0
  exec-timeout 5 0
  password ciscoconpass
  logging synchronous
  login
!
line aux 0
!
line vty 0 4
  exec-timeout 5 0
  password ciscovtypass
  login
  transport input telnet
!
```

Yes, we can read the console, aux and vty passwords because we used enable password which stores password in plain text.

- f. Repeat the configuration portion of steps 3a through 3f on router R2.

Step 4: Encrypt clear text passwords.

- a. Use the **service password-encryption** command to encrypt the console, aux, and vty passwords.

R1(config)# **service password-encryption**

- b. Issue the **show run** command. Can you read the console and vty passwords? Explain.

[Your Answer Here]

```
.
line con 0
  exec-timeout 5 0
  password 7 0822455D0A1606181C1B0D1739
  logging synchronous
  login
!
line aux 0
!
line vty 0 4
  exec-timeout 5 0
  password 7 0822455D0A1613030B1B0D1739
  login
  transport input telnet
,
```

No, since we used service password-encryption command to encrypt console, aux and vty password we cannot read them.

At what level (number) is the default enable secret password encrypted? _____ 5 _____

At what level (number) are the other passwords encrypted? _____ 7 _____

Which level of encryption is harder to crack and why?

Level 5 encryption is harder to since it is a one way hash and cannot be reversed while level 7 encryption is reversible and can be cracked with known tools.

Task 2: Configure a Login Warning Banner on Routers R1 and R2.

Step 1: Configure a warning message to display prior to login.

- a. Configure a warning to unauthorized users with a message-of-the-day (MOTD) banner using the **banner motd** command. When a user connects to one of the routers, the MOTD banner appears before the login prompt. In this example, the dollar sign (\$) is used to start and end the message.

```
R1(config)# banner motd $Only YOURNAME permitted!$
```

```
R1(config)# exit
```

- b. Issue the **show run** command. What does the \$ convert to in the output?

```
!  
banner motd ^COnly Shaan permitted!^C  
!
```

The \$ is not shown in the text. It is used as a marker to start and end the text.

Task 3: Configure Enhanced Username Password Security on Routers R1 and R2.

Step 1: Create a new user account with a secret password.

- a. Create a new user account with an encrypted password.

```
R1(config)# username user01 secret user01pass
```

- b. Exit global configuration mode and save your configuration.
- c. Display the running configuration.


```
R1#show run
Building configuration...

Current configuration : 1264 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
security passwords min-length 10
!
hostname R1
!
!
!
enable secret 5 $1$mERr$WvpW0n5HghRrqnrxXCUU1.
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username user01 secret 5 $1$mERr$AAEJVg9OAgW/Zn59wtI4d/
!
!
license udi pid CISCO2911/K9 sn FTX1524H25L-
!
!
!
!
!
!
!
!
!
!
no ip domain-lookup
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
```

```

!
interface GigabitEthernet0/2
  no ip address
  duplex auto
  speed auto
  shutdown
!
interface Serial0/3/0
  ip address 10.1.1.1 255.255.255.252
  clock rate 64000
!
interface Serial0/3/1
  no ip address
  clock rate 2000000
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
router rip
  network 10.0.0.0
  network 192.168.1.0
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^COnly Shaan permitted!^C
!
!
!
!
line con 0
  exec-timeout 5 0
  password 7 0822455D0A1606181C1B0D1739
  logging synchronous
  login
!
line aux 0
!
line vty 0 4
  exec-timeout 5 0
  password 7 0822455D0A1613030B1B0D1739
  login
  transport input telnet
!
!
!
end

```

Step 2: Test the new account by logging in to the console.

- a. Set the console line to use the locally defined login accounts.

```
R1(config)# line console 0
R1(config-line)# login local
R1(config-line)# end
R1# exit
```

- b. Exit to the initial router screen which displays: R1 con0 is now available, Press RETURN to get started.
- c. Log in using the previously defined username **user01** and the password **user01pass**.

What is the difference between logging in at the console now and previously?

The difference between logging in at the console now is that it asks for username and password whereas previously it only asked for line console password.

- d. After logging in, issue the **show run** command. Were you able to issue the command? Explain.

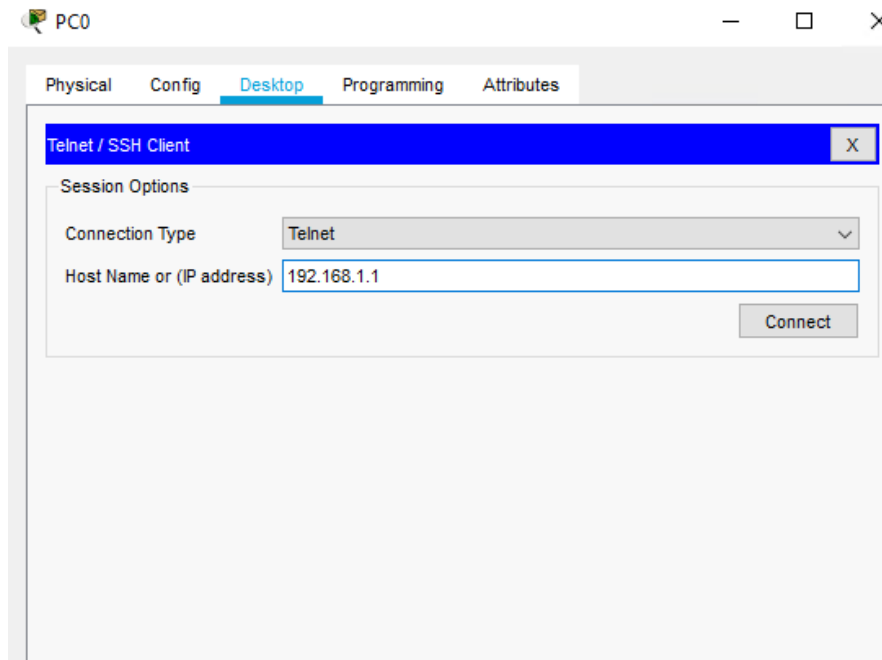
No, I couldn't issue the show run command since we need to enter privileged EXEC mode to issue show run command.

- e. Enter privileged EXEC mode using the **enable** command. Were you prompted for a password? Explain.

Yes, the password was prompted since privileged EXEC mode is protected.

Step 3: Test the new account by logging in from a Telnet session.

- a. Open "Desktop" and Telnet/SSH Client and choose "Telnet".
- b. Input the R1 G0/0 IP address **192.168.1.1** in the **Host Name (or IP address)** field.



Were you prompted for a user account? Explain.

No, I was not prompted for a user account since vty lines only had a line password configured not individual user accounts.

- c. Set the vty lines to use the locally defined login accounts.

```
R1(config)# line vty 0 4
```

```
R1(config-line)# login local
```

- d. From PC-A, telnet to R1 again.

Were you prompted for a user account? Explain.

```
Username: user01
Password:
R1>enable
Password:
R1#
```

Yes, It prompted me for a user account since the vty lines were configured for local authentication using login local.

- e. Log in as **user01** with a password of **user01pass**.
- f. During the Telnet session to R1, access privileged EXEC mode with the **enable** command.

What password did you use?

cisco12345

Task 4: Configure the SSH Server on Router R1 and R2.

In this task, use the CLI to configure the router to be managed securely using SSH instead of Telnet. Secure Shell (SSH) is a network protocol that establishes a secure terminal emulation connection to a router or other networking device. SSH encrypts all information that passes over the network link and provides authentication of the remote computer. SSH is rapidly replacing Telnet as the remote login tool of choice for network professionals.

Note: For a router to support SSH, it must be configured with local authentication, (AAA services, or username) or password authentication. In this task, you configure an SSH username and local authentication.

Step 1: Configure a domain name.

Enter global configuration mode and set the domain name.

```
R1# conf t
```

```
R1(config)# ip domain-name networksecurity.com
```

Step 2: Configure a privileged user for login from the SSH client.

- Use the **username** command to create the user ID with the highest possible privilege level and a secret password.

```
R1(config)# username admin privilege 15 secret cisco12345
```

- Exit to the initial router login screen. Log in with the username admin and the associated password. What was the router prompt after you entered the password?

```
User Access Verification
```

```
Username: admin
```

```
Password:
```

```
R1#
```

The user is in the privileged EXEC mode after the password is entered. Since the admin is given highest level privilege.

Step 3: Configure the incoming vty lines.

Specify a privilege level of **15** so that a user with the highest privilege level (15) will default to privileged EXEC mode when accessing the vty lines. Other users will default to user EXEC mode. Use the local user accounts for mandatory login and validation and accept only SSH connections.

```
R1(config)# line vty 0 4
```

```
R1(config-line)# privilege level 15  
R1(config-line)# login local  
R1(config-line)# transport input ssh  
R1(config-line)# exit
```

Note: The **login local** command should have been configured in a previous step. It is included here to provide all commands, if you are doing this for the first time.

Note: If you add the keyword **telnet** to the **transport input** command, users can log in using Telnet as well as SSH, however, the router will be less secure. If only SSH is specified, the connecting host must have an SSH client installed.

Step 4: Erase existing key pairs on the router.

```
R1(config)# crypto key zeroize rsa
```

Note: If no keys exist, you might receive this message: % No Signature RSA Keys found in configuration.

Step 5: Generate the RSA encryption key pair for the router.

The router uses the RSA key pair for authentication and encryption of transmitted SSH data.

Step 6: Configure the RSA keys with 1024 for the number of modulus bits. The default is 512, and the range is from 360 to 2048.

```
R1(config)# crypto key generate rsa general-keys modulus 1024
```

The name for the keys will be: R1.networksecurity.com

% The key modulus size is 1024 bits

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

```
R1(config)#
```

*Dec 16 21:24:16.175: %SSH-5-ENABLED: SSH 1.99 has been enabled

- a. Issue the **ip ssh version 2** command to force the use of SSH version 2.

```
R1(config)# ip ssh version 2
```

```
R1(config)# exit
```

Step 7: Verify the SSH configuration.

- a. Use the **show ip ssh** command to see the current settings.

```
R1# show ip ssh
```

```
R1#show ip ssh
```

```
SSH Enabled - version 2.0
```

```
Authentication timeout: 120 secs; Authentication retries: 3
```

```
R1#
```

- b. Fill in the following information based on the output of the **show ip ssh** command.

SSH version enabled: version 2.0

Authentication timeout: 120 seconds

Authentication retries: 3

Step 8: Configure SSH timeouts and authentication parameters.

The default SSH timeouts and authentication parameters can be altered to be more restrictive using the following commands.

```
R1(config)# ip ssh time-out 90
```

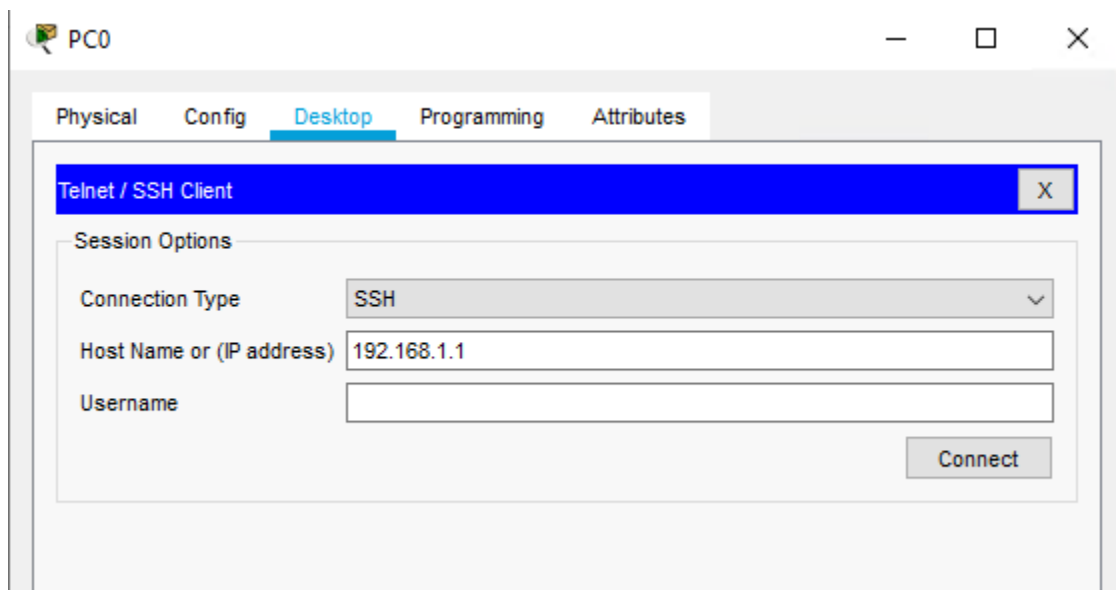
```
R1(config)# ip ssh authentication-retries 2
```

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
R1#
```

Task 5: Configure the SSH Client.

Step 1: Verify SSH connectivity to R1 from PC-A.

- Launch “Telnet/SSH Client” from Desktop.
- Input the R1 G0/0 IP address **192.168.1.1** in the **Host Name (or IP address)** field.



- Enter the **admin** username and password **cisco12345** when prompted

```
192.168.1.1 - PuTTY
login as: admin
Using keyboard-interactive authentication.
Password:
Unauthorized access strictly prohibited!
R1#
```

- d. At the R1 privileged EXEC prompt, enter the **show users** command.

R1# **show users**

What users are connected to router R1 at this time?

```
R1#show users
      Line      User      Host(s)      Idle      Location
  0 con 0      admin      idle        00:01:12
*390 vty 0      admin      idle        00:00:00

Interface      User      Mode      Idle      Peer Address
R1#
```

Admin is connected to router R1 currently via vty line.

- e. Try to open a Telnet session to your router from PC-A. Were you able to open the Telnet session? Explain.

No, The vty lines only allow ssh connection since they are configured to transport input ssh which blocks telnet.

- f. Open a SSH session to the router from PC-A. Enter the **user01** username and password **user01pass** in the SSH window to try connecting for a user who does not have privilege level of 15.

If you were able to login, what was the prompt?

Yes, The motd was prompted. The user01 started from user EXEC mode since it didn't have level 15 privilege.

- g. Use the **enable** command to enter privilege EXEC mode and enter the enable secret password **cisco12345**.

```
Password:
Only Shaan permitted!
R1>en
Password:
R1#
```